MODULE 2 — Data Cleaning: Noise, Bias, and Tradeoffs

Bias Shaping, Not Fixing Data

Why This Module Matters

Humans think of cleaning as **removing errors**. AI thinks of cleaning as **shaping bias**.
Every cleaning choice:

- changes what patterns are visible
- suppresses some behaviors
- amplifies others
- alters downstream decisions

There is no "neutral" cleaning.

Anchor Back to Our Decision (Always)

From Modules 0–1:
We are deciding which customer cohorts should receive churn interventions under budget and operational constraints, with uncertainty about whether recent behavioral changes reflect true churn risk or temporary noise.
Cleaning does **not** make data "better."It makes data **more aligned with one interpretation of risk**.

Core Reframe (Memorize This)

To AI, data cleaning is not error removal — it is intentional bias shaping under uncertainty.
If you don't choose the bias, the system will — implicitly.

Takeaways

- Every data decision made introduces bias into the system; therefore, cleaning does not make data *better*.

- Focus of data cleaning then becomes how to mitigate understood risk about a sought-after decision for a given project.
  - "Data cleaning exists to mitigate *decision-relevant risk*, not to maximize data "quality" in the abstract.
  - "Data is never cleaned *globally*, it is cleaned **relative to a decision**."

Why This Is True (AI-Native View)

Every decision has:

- different failure costs
- different tolerance for uncertainty
- different asymmetries (false positives vs false negatives)
- different notions of "harm"

Because of that:

The same data can be "correctly cleaned" in two incompatible ways — depending on the decision it supports.

AI systems reason this way naturally; humans have to learn it explicitly.

Concrete Illustration (Same Data, Two Decisions)

Same data source

Customer usage logs

Decision X — *"Who should receive costly retention incentives?"*

- Risk to mitigate: **false positives**
- Cleaning bias:
  - smooth short-term volatility
  - be conservative about anomalies
- Accepts missing signals to avoid waste

Decision Y — *"Who needs immediate human outreach?"*

- Risk to mitigate: **false negatives**
- Cleaning bias:
  - preserve extreme drops
  - treat missing data as warning signals
- Accepts noise to avoid missed churn

Same data. Opposite cleaning choices. Both are *correct* — for their decisions.

Takeaways

- The way *false positives & false negatives* are talked about in this section is very different from how these concepts are conventionally taught.
  - Typically, this is taught as how to interpret predictions of a classification model -> what are we more worried about? Type I or Type II error here?
  - This section suggests that this is actually something that needs to be frame *before ANY data cleansing begins*.
  - **Benefit:** EDA outcomes & modeling outcomes become easier to explain, because the entire pipeline will be cohesive in what assumptions were used in determining what decisions were made at each step.

Why "Neutral Cleaning" Is a Myth

If you try to clean data to be:

- maximally smooth
- maximally complete
- maximally symmetric

You end up optimizing for **no decision at all** — which is how silent failure enters systems.
AI will then confidently act on data that was never aligned with its goal.

AI's Cleaning Decision Questions

Before changing a single value, AI implicitly asks:

1. **Is this error random, systematic, or structural?**
2. **Does this cleaning increase signal or destroy it?**
3. **What downstream behavior does this change?**
4. **Who benefits or loses from this transformation?**

If you cannot answer these, cleaning is premature.

Q: How is *error* defined in the context of this module?

Appendix

- Under Module 2, see *Question: How is Error Being Defined In This Module?*

A: **"Error" means any deviation between the data you observe and the data-generating process that would be ideal for the decision you are trying to make.**

- **In this module, "error" means any deviation from ideal evidence — and the first question is whether that deviation is accidental, biased, or intrinsic to the phenomenon itself.**
- That includes:
  - missing values
  - outliers
  - mislabeled outcomes

    - delayed labels
    - distorted measurements
    - censored observations

A: Three types of error, *random, systematic, & structural*

- Random Error - noise introduced by chance, measurement imprecision, or transient glitches.
    - **Key property** - error is **independent** of the outcome you care about.
    - **Cleaning implication** - removing/smoothing often **reduces variance without biasing decisions**.
    - **Danger** - mistaking systematic or structural error for random.
- Systematic Error - error that occurs **consistently in one direction or subgroup**.
    - **Examples**
        - Churn labels recorded late for long-tenure customers
        - Usage undercounted for mobile users
    - **Key property** - error is **correlated with the decision-relevant outcome**.
    - **Cleaning implication** - naive "fixing" can **hide bias** rather than remove it.
    - **Danger** - treating systematic error as random or smoothing it
- Structural Error - error caused by **how the system itself is designed**, not by noise.
    - **Examples**
        - Churn only recorded when an account is formally closed
        - Disengaged users stop generating usage data (MNAR)
        - Labels encode business rules, not customer intent
        - Outliers represent threshold-triggered behavior
    - **Key property** - error is inseparable from the phenomenon being studied
    - **Cleaning implication** - "Fixing" it **destroys the signal you care about**.
    - **Danger** - cleaning the data into lying.

Common Cleaning Fallacies (Silent Failure Patterns)

- Treating missing data as accidental when it's meaningful
- Removing outliers that represent *real churn behavior*
- "Fixing" labels to match expectations
- Enforcing symmetry where the real world is asymmetric
- Making data look nicer at the cost of truth

Clean-looking data can be **more dangerous** than messy data.

Missingness Is a Signal (Not a Bug)

AI distinguishes missingness types conceptually:

- **MCAR** — Missing Completely At Random(rare, safest assumption)
- **MAR** — Missing At Random *conditional on other variables*(common, risky if misunderstood)
- **MNAR** — Missing Not At Random(very common in churn: disengaged users stop generating data)

🚨 Treating MNAR as MCAR is one of the most common DS failures.

Outliers: Defects or Warnings?

In churn contexts:

- extreme drops in usage
- sudden billing changes
- bursty support activity

These may be:

- logging glitches ❌
- or *exactly* the behavior you care about ✅

AI will not know which unless you tell it.

One-Sentence Mental Model (Worth Memorizing)

Data cleaning is the act of choosing which kinds of mistakes you are willing to make, in service of a specific decision.

That sentence will keep you out of more trouble than any checklist.

Prompt Engineering Pattern (Critical)

When involving AI in cleaning, use:

"Propose multiple cleaning strategies, explain their assumptions, and identify when each would be dangerous."

Never ask:

"Clean this dataset."

That invites overconfidence.

🧪 Module 2 Mini-Exercise (Key Learning Moment)

We'll reuse the **historical churn labels** you classified as **heuristic**.

Step 1 — Ask for Conflicting Strategies (Conceptually)

Imagine two plausible cleaning strategies for churn labels:

- Strategy A: *Aggressively clean noise*
- Strategy B: *Preserve ambiguity*

You don't need code — just intent.

Step 2 — For Each Strategy, Answer:

For **each** strategy:

1. What assumption does it make about churn?
2. What bias does it introduce?
3. When would it be **dangerous**?

Think like a reviewer, not an implementer.

Rules (Important)

❌ No modeling
❌ No metrics
❌ No "best practice" language
✅ Explicit tradeoffs only

Why This Module Is Hard (And Valuable)

Junior practitioners ask:
"What's the right way to clean this?"
Senior practitioners ask:
"Which wrong assumptions am I willing to live with?"
AI-native systems require the second mindset.

Relevant Articles

Bias-Variance Tradeoff

- https://en.wikipedia.org/wiki/Bias%E2%80%93variance_tradeoff?utm_source=chatgpt.com

Missingness In Semi-Supervised Learning

- https://www.researchgate.net/figure/Schematic-illustration-showing-how-MCAR-MAR-and-MNAR-missingness-mechanisms-influence_fig1_398357316
- Untitled Attachment

Outlier Analysis: Identification, Impact, and Comprehensive Treatment Strategies

- https://medium.com/%40silva.f.francis/outlier-analysis-identification-impact-and-comprehensive-treatment-strategies-f547c08c36c5

Choosing the Right Data Preprocessing Technique: A Decision Tree Approach

- https://www.statology.org/choosing-the-right-data-preprocessing-technique-a-decision-tree-approach/?utm_source=chatgpt.com